3.5 Describe an octal number in expanded notation

The octal number system is a base 8 number system. It uses eight digits: 0, 1, 2, 3, 4, 5, 6, and 7.

A subscript 8 identifies a number as octal. For example, $2\ 5\ 3_8$ and 1372_8.

Like the decimal and binary number systems, the octal system uses place values. However, each place value is a power of 8 rather than a power of 10 or 2.

3.6 Express an octal number in expanded notation

Place Value

From right to left, the place values are ones, eights, sixty-fours, five hundred twelves, and so on.

… , 512 , 64 , 8 , 1

Each place value is a power of 8:

Power of 8	Place Value
8^0	1
8^1	8
8^2	64
8^3	512
8^4	4096

In exponent form, the place values are

$$\ldots 8^4, 8^3, 8^2,\ 8^1,\ 8^0$$

Example

Express 253_8 in expanded notation.

3.9 Convert whole numbers using the following systems:

3.9.1 Decimal

3.9.2 Binary

3.9.3 Octal

Octal to Decimal Conversions

To convert an octal number to decimal:

- Multiply each digit by its corresponding power of 8.
- Add the results.

Convert 253_8 to decimal.

Convert 1372_8 to decimal.

Converting Decimal to Octal

To convert a decimal number to octal:

- Divide the decimal number by 8.
- Record the remainder.
- Continue dividing the quotient by 8 until the quotient is 0.
- Read the remainders from bottom to top to obtain the octal number.

Convert 156_{10} to octal.

Convert 231_{10} to octal form.

Convert the following decimal numbers to octal form.

a) 311

b) 699

c) 1875